Held-out Accuracy Is Not New-Collection Accuracy: A Cross-Collection Benchmark of Poultry Dropping Classifiers and a Fault-Isolated Pen-Camera Detector on a QNX Raspberry Pi 5

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Abstract

Deep networks classify photographs of poultry droppings into healthy, coccidiosis, salmonella and Newcastle disease with reported accuracies of 98 to 99%, but these numbers come from random splits of a single collection. We retrained the classifiers earlier work used (MobileNetV2 and EfficientNet-B0) and our YOLO11n detector on the same split of the public Tanzanian farm photographs. We then tested all of them on the same 679 held-out farm photographs and on 1,255 photographs from a separate collection whose labels were confirmed by PCR. Every model scores 98 to 99% on the farm test and 75 to 78% on the lab-confirmed photographs when it always gives an answer, a drop of more than 20 points, so the test set matters more than the architecture. An audit of the split found 49 test photographs (7.2%) that are exact copies of training photographs. Removing them changes the farm accuracy by only 0.1 points, so they do not explain the gap. The detector's box score ranks photographs well but is not a calibrated probability. We also graded every lab-confirmed photograph for appearance with three AI raters (Krippendorff's alpha 0.76). Salmonella droppings look almost healthy, yet both models still name them correctly, while normal looking coccidiosis is mostly missed. Since a fixed pen camera sees each dropping at 8 to 30 pixels, we fine-tuned the detector on public barn video with real droppings pasted onto the litter, split by camera. On unseen cameras, recall rises from 0.29 to 0.80 and precision from 0.11 to 0.93, at a cost of 0.02 mAP50 on close-ups. The full pipeline runs on a Raspberry Pi 5 under the QNX microkernel. Each stage is an isolated process, and any stage we killed was restarted in 20 to 33 ms while the others kept running. We release the code, the per-photograph predictions and every figure with its data.

1 Introduction

Coccidiosis alone was estimated to cost the global chicken industry about £10.4 billion at 2016 prices [5]. Coccidiosis, salmonellosis and Newcastle disease change the colour and consistency of droppings, and farmers already look at droppings to judge flock health. This has made dropping photographs a popular target for deep learning. On the public Tanzanian collection of Machuve et al. [24, 25], convolutional classifiers reach 98 to 99% accuracy [10, 11, 17, 25, 29], and several groups have proposed phone, web or edge systems around them [10, 11, 30].

These accuracies come from random splits of a single collection. A random split by photograph places pictures from the same farm visit, and sometimes the same picture, on both sides of the split [2, 19]. The original authors wrote that they did not expect their model to be affected by distribution shift [25]. This has not been tested for the four-class task. The same group released a second collection whose labels were confirmed by PCR [23], which offers a direct test. A second gap is the camera. Published systems classify close-up photographs taken by a person. A camera fixed above a pen sees each dropping at 8 to 30 pixels, among birds, feeders and litter, and nobody is there to restart the device when software fails.

We address both gaps with one detector and one honest protocol. Our contributions are as follows.

1. A benchmark of the classifiers earlier work used (MobileNetV2, EfficientNet-B0), retrained on the same split as our YOLO11n detector and tested on the same 679 farm photographs and on 1,255 lab-confirmed photographs that no model saw. Every model scores 98.5 to 99.6% on the farm test, in line with published numbers, and 74.6 to 78.2% on the lab-confirmed photographs. Our detector is not a better classifier than the earlier models, and the drop is the same for all of them. To our knowledge this is the first such test for the four-class task.
2. An audit of the split. We find that 49 of the 679 test photographs (7.2%) are exact copies of training photographs, and we show that removing them changes the farm accuracy by only 0.1 points. We also show that the detector's box score is not a calibrated probability.
3. A pen-scale fine-tune trained with real droppings pasted onto barn video frames, split by camera. On cameras it never saw, recall rises from 0.29 to 0.80 and precision from 0.11 to 0.93, at a cost of 0.02 mAP50 on close-ups.
4. A complete pipeline running on a Raspberry Pi 5 under the QNX microkernel, with every model timed on the device. Each stage is an isolated process. When we kill any stage, it is restarted in 20 to 33 ms while the oth-

ers keep running. To our knowledge no peer-reviewed work has run poultry health vision on a microkernel real-time operating system.

We release the code, the per-photograph predictions, the duplicate hashes and every figure with the table behind it.

2 Related work

Classifying poultry droppings. Machuve et al. collected the Tanzanian photographs we use and compared five convolutional networks [25]. With fine-tuning, MobileNetV2 reached 98.02% and Xception 98.24% on a farm-labelled test set, and the authors recommended MobileNetV2 for deployment. Later work on the same photographs or their 224-pixel Kaggle copy reports similar or higher numbers. Degu and Simegn combined a YOLOv3 detector with a ResNet-50 classifier for a phone application and report 98.7% [10]. Reyad et al. report 99.25% with a majority-vote ensemble [29], and Huda et al. 97.78% with a hybrid of EfficientNetV2-B0 and DenseNet-121 [17]. The closest pipeline to ours is that of Dhungana et al., a YOLO11n detector followed by an EfficientNet-B0 classifier, with a detector mAP50 of 0.881 and a classifier accuracy of 99.12% [11]. Sakib et al. capture photographs with an ESP32-CAM and classify them with EfficientNet-B0 on a server, reporting 99.19% on a public test set and 95.62% on a separate local-farm set [30]. Other groups built their own data. Aworinde et al. released 14,618 cleaned Nigerian smartphone photographs at 100×100 pixels [1], and Nakrosis et al. classified 487 Lithuanian photographs of droppings on litter into six appearance classes, with 91.78% accuracy [26]. Wang et al. detected droppings on a conveyor belt below caged broilers [33]. A survey by Li lists the open poultry vision datasets [22].

Testing beyond one collection. Bodempudi et al. are the closest to our evaluation. They classified droppings as *Salmonella* PCR positive or negative across United States and African images and report up to 85% across regions [6]. Their task is binary and their test set is a different region. We test all four classes, on a collection from the same group whose labels were confirmed in the laboratory, which separates the effect of label source and collection from the effect of region. More broadly, leakage between training and test data has inflated results across many fields [19], near-duplicate images between the splits of standard benchmarks have been documented [2], and networks often learn shortcuts such as framing or background that stand in for the class [13]. We use the calibration and selective prediction tools of Guo et al. [15] and Geifman and El-Yaniv [12].

Edge deployment and operating systems. Livestock vision on small devices usually means shrinking the model, for example by up to 270 times for microcontrollers [35]. Published dropping systems run on a phone, a server or a laptop [10, 11, 30]. None that we found addresses what happens when part of the software fails on an unattended device. Microkernels move drivers and services into isolated processes, so a fault stays in one process [3, 20]. A vendor white paper compares QNX 8 with real-time Linux on a Raspberry Pi 5 with a neural accelerator and reports similar throughput with less variation [4]. It is not peer reviewed and does not measure fault recovery. We measure recovery directly.

3 Methods

3.1 Data

Farm photographs. We used the poultry faecal image collection of Machuve et al. released on Zenodo (record 4628934, CC BY 4.0) [24, 25]. It contains 6,812 full-resolution smartphone photographs taken on farms in Arusha and Kilimanjaro, Tanzania, with official bounding boxes in YOLO format for four classes: healthy, coccidiosis, salmonella and Newcastle disease. We downloaded the archives through the Zenodo API and checked each against the MD5 checksum the publisher lists. Following the dataset, we refer to the disease classes as *coccidiosis-like*, *salmonella-like* and *Newcastle-like*, because the farm labels describe the appearance of the dropping and the farm a sample came from, not a laboratory diagnosis of each dropping.

Lab-confirmed photographs. The same group released a second, separate collection of 1,255 photographs whose labels were confirmed by polymerase chain reaction (PCR) (Zenodo record 5801834) [23]. It contains 373 coccidiosis, 347 healthy, 349 salmonella and 186 Newcastle disease photographs. No model in this study was trained, tuned or selected on any of these photographs. We use them only as an external test set.

Barn video. To teach the detector the size droppings have in a fixed pen camera, we took background frames from two public video collections of broiler houses: the PIO videos (Zenodo record 16686320, CC BY 4.0) [7, 8] and the NESTLER poultry videos (Zenodo record 15063494, CC BY 4.0) [27]. No dropping in these videos is labelled, and we did not label any.

3.2 Split

The farm photographs were split 80/10/10 by photograph within each class, using a fixed seed on the sorted file names, giving 5,451 training, 679 validation and 679 test photographs. This is the kind of random split earlier work on this collection reports. The detector and every baseline in Section 3.5 use exactly this split. To measure

leakage, we reduced every photograph to a 256-bit difference hash [21] and recorded, for each test photograph, the Hamming distance to the closest training photograph.

3.3 Detector

Our detector is YOLO11n [18] with four classes, fine-tuned from COCO weights at 640 pixels with early stopping (patience 20, seed 0) on a single laptop GPU odoconfirm the GPU model: the run notes say RTX 3060. The Newcastle-like class has about a sixth of the photographs of the others, so its training photographs are repeated three times, after the split and in the training set only. Training stopped at epoch 69 and the weights of epoch 49, the best on validation, were kept (`dropping4-laptop`). We exported it to ONNX at 320, 416 and 640 pixels.

To compare the detector with photograph classifiers, we score a whole photograph with it: the photograph is letterboxed to 640 pixels, every candidate box keeps its class scores, and each class takes its best score in the photograph. The photograph's call is the class with the highest score, or *no detection* when no box reaches 0.25, the confidence threshold the deployed pipeline uses.

3.4 Pen-scale fine-tuning

In a pen camera at 1280×720 a dropping covers 8 to 30 pixels, against a median of 12 to 22% of the photograph for the disease classes in the farm collection. We built a pen-scale training set by copy-paste augmentation [14]: real labelled droppings were cut from the farm photographs along their official boxes, shrunk to pen size, and pasted onto empty litter in barn video frames. A dropping from a training photograph is only ever pasted into a training tile, and likewise for validation and test, so the farm split carries over. Background frames were split *by camera*, never by frame. Five PIO videos and the remaining NESTLER cameras form the training set, one NESTLER camera the validation set, and three NESTLER cameras plus the PIO video used in our live demonstration the test set. Frames with birds and equipment and no pasted dropping were kept as negatives. The model `dropping4-pen` was fine-tuned from `dropping4-laptop` on these tiles mixed with the original farm photographs, for up to 40 epochs (patience 10). It stopped at epoch 38.

We state the limitation plainly: the pen test is synthetic. It measures whether the model finds a real dropping of pen size on litter and cameras it has never seen, not field accuracy on naturally occurring droppings.

3.5 Baselines

Published results on this collection come from different splits, image sizes and protocols, so we retrained the classifiers earlier work used instead of quoting their numbers. MobileNetV2 [31] is the model Machuve et al. recommended for deployment [25], EfficientNet-B0 [32] is the classifier stage of a recent YOLO11 and EfficientNet-B0 pipeline [11]. Each was initialised from ImageNet weights and trained on the same 5,451 training photographs, resized to 224×224 without keeping the aspect ratio (as in the 224-pixel copy of the collection most earlier work used), with AdamW and a one-cycle learning rate schedule. MobileNetV2 was trained for 20 epochs and EfficientNet-B0 for 8, both with one seed, because of the compute time available to us. Augmentation was random crops, horizontal and vertical flips, and small brightness and contrast changes. We did not change hue, because colour is the main visual sign of disease. The epoch with the best validation accuracy was kept. Both reached over 98% validation accuracy within four epochs, so the shorter EfficientNet-B0 run is unlikely to change the comparison, but a single seed does not measure run-to-run variation.

3.6 Evaluation

All models are scored on the same 679 farm test photographs and the same 1,255 lab-confirmed photographs. We report accuracy, per-class recall and macro-F1. Proportions carry 95% Wilson score intervals [34], which behave well near 0 and 1. For the detector we also report a risk-coverage curve [12], the accuracy of the answers when the model only answers above a confidence threshold, and a reliability diagram with the expected calibration error over ten equal-width bins [15].

3.7 Appearance grades for the lab-confirmed photographs

PCR says whether a pathogen is in the sample, not how the dropping looks. To ask whether the models fail on droppings that look normal, we graded every one of the 1,255 lab-confirmed photographs for appearance. A written rubric scores six visual variables 0, 1 or 2, or leaves them empty when the photograph does not show them: form, the white urate cap, colour, blood, mucus or foam, and undigested feed. These are the signs used in clinical checks of droppings. The grade is computed, never chosen, as $\max(1, 10 - p)$ where p is the sum of the six scores with blood counted twice, so every lost point has a named reason. Grades 9 and 10 mean the dropping looks normal.

The raters were three AI vision models, not people: Claude Opus, Claude Sonnet and Claude Haiku, each run with the same written instructions and no shared context. Each photograph got a random identifier and was re-encoded at 768 pixels with no metadata, and the key linking identifiers to file names and PCR classes stayed outside the raters' folder until grading ended. Each rater saw the photographs in its own random order. 126 photographs (10%) were shown a second time under a new

identifier to measure each rater's agreement with itself. The released sub-score for a photograph is the median of the raters who could judge it, needing at least two, and its grade is recomputed from those medians. We report Krippendorff's alpha for ordinal data across the three raters [16] and quadratic weighted kappa for each pair and for each rater against itself [9].

3.8 Deployment on QNX

The deployed system runs on a Raspberry Pi 5 with a Camera Module 3 under QNX 8.0, a microkernel real-time operating system [3]. The pipeline is split into separate processes (sections), each with its own memory and a fixed real-time priority: camera capture in C, a still-spot finder that flags new objects that appeared and stayed still, the detector, an explanation step, and links to the dashboard and a wrist-worn display. Full frames stay in a shared-memory ring of four slots, and sections exchange only small messages through QNX native message passing. A supervisor restarts any section that exits. Inference uses ONNX Runtime [28] 1.24.4 from the QNX open-source package repository, called from a small C bridge because the repository's Python binding does not load on QNX (Section 5). We report the detector's inference time on the Pi with four threads, without image decoding or post-processing.

4 Results

4.1 All models drop on lab-confirmed photographs

Table 1 and Figure 1 compare every model on the two test sets. On the farm test all of them are in the range earlier work reports. MobileNetV2 calls 99.6% of the 679 photographs correctly (95% CI 98.7 to 99.8%), EfficientNet-B0 99.0% (97.9 to 99.5%), and our detector 98.5% (97.3 to 99.2%) when it always gives its best guess. On the 1,255 lab-confirmed photographs every model loses more than 20 points. MobileNetV2 falls to 74.6% (72.1 to 76.9%), EfficientNet-B0 to 78.2% (75.9 to 80.4%) and the detector to 76.3% (73.9 to 78.6%). For each model the intervals of the two test sets are far apart, so the drop is not sampling noise.

Our detector does not classify better than the classifiers earlier work used. On the lab-confirmed photographs the intervals of all three models overlap, and EfficientNet-B0 has the highest point estimate. The models fail on different classes. The classifiers keep far more Newcastle-like photographs (recall 0.70 for MobileNetV2 and 0.88 for EfficientNet-B0, against 0.53 for the detector), while the detector keeps more healthy ones (0.81 against 0.66 and 0.68). The per-class differences are large, but the differences in overall accuracy are small next to the drop that every model shows. Under this protocol the published accuracies of 98 to 99% describe photographs like the training photographs, and the choice of architecture matters less than the choice of test set.

Table 1: Accuracy and macro-F1 on the held-out farm test (679 photographs) and the lab-confirmed photographs (1,255), for models trained on the same 5,451 photographs. One training run per model. The detector is scored with its best class in each photograph (always answers) and with the deployed 0.25 confidence threshold (below it the photograph counts as wrong).

	Farm test		Lab-confirmed	
Model	Accuracy	Macro-F1	Accuracy	Macro-F1
MobileNetV2 [25]	99.6	99.7	74.6	75.3
EfficientNet-B0 [11]	99.0	98.6	78.2	79.5
YOLO11n (ours), always answers	98.5	98.3	76.3	75.1
YOLO11n (ours), 0.25 threshold	97.9	97.7	67.1	70.3

4.2 Where the errors go

Figure 2 shows the detector's calls on both sets. On the farm test the diagonal holds 95 to 100% of every class. On the lab-confirmed photographs most errors are not confusions between diseases. They are photographs with no detection at all (20% of healthy and 27% of Newcastle-like photographs) and photographs called healthy or salmonella-like by mistake. Newcastle-like recall falls the most, from 94.6% (35 of 37) to 41.9% (78 of 186). This class has 376 photographs in the whole farm collection, against more than 2,000 for each of the others (Figure 3).

4.3 Duplicates do not explain the farm score

Forty-nine of the 679 farm test photographs (7.2%) have a training photograph with an identical 256-bit difference hash, and inspection of the closest pairs shows they are the same image saved under two file names (Figure 4). Every other test photograph differs from its nearest training photograph in at least 40 bits. The detector calls all 49 copies correctly. Without them its farm accuracy is 97.8% (95% CI 96.3 to 98.7%), against 97.9% with them. The copies are a flaw in the split, but they are not why the farm test overstates performance on new photographs.

4.4 The box score ranks well but is not a probability

If the detector only answers when its best box clears a threshold, its answers become more reliable (Figure 5). On the lab-confirmed photographs, raising the threshold from 0.25 to 0.80 raises the accuracy of the answers from 78.8% to 87.4%, but the detector then answers only 27.8% of photographs. Its scores are also under-confident on both sets (Figure 6). The expected calibration error is 0.177 on the farm test and 0.172 on the lab-confirmed photographs, and in both cases the photograph is right

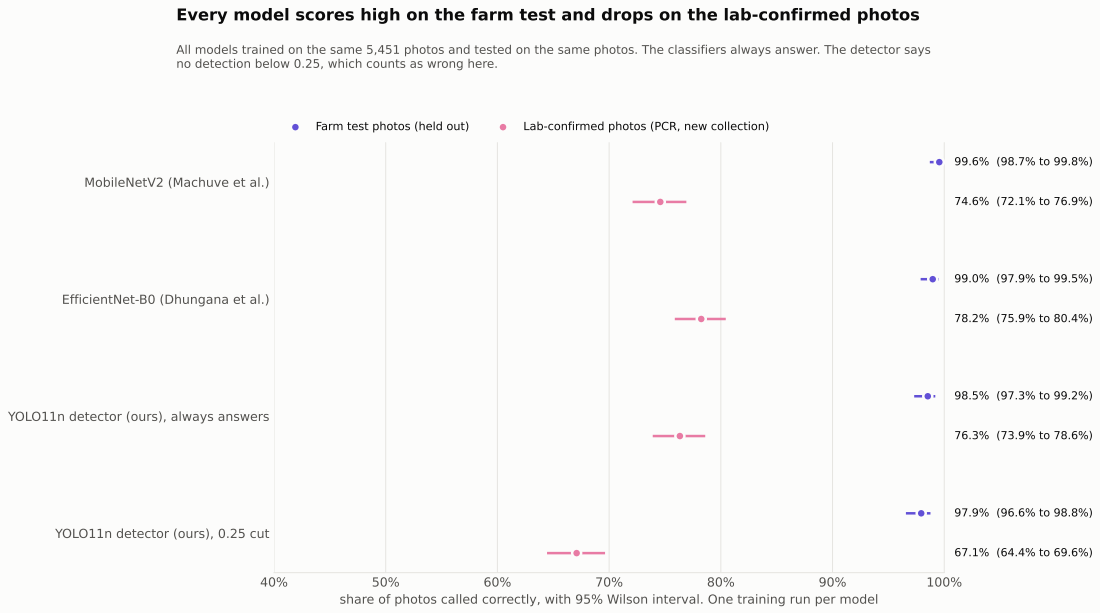


Figure 1: Accuracy of each model on the farm test and on the lab-confirmed photographs. One training run per model. Values are written beside each mark.

more often than the score says. The score should therefore be calibrated, for example by temperature scaling [15] on the validation split, before it is shown to a farmer as a percentage.

4.5 What the lab-confirmed droppings look like

The three raters agreed well on the grade, with Krippendorff's alpha 0.76 over the 1,255 photographs and pairwise weighted kappa from 0.71 to 0.84. Agreement on single variables was lower, from 0.56 for colour and 0.57 for undigested feed to 0.75 for the urate cap and 0.74 for blood. Each rater agreed with itself on the 126 repeats with weighted kappa 0.71 to 0.86, but gave the identical grade only 37 to 47% of the time, so a single grade is precise to about one point. Nine photographs showed no dropping that could be judged, 331 were only partly visible, and on 234 the raters' grades spread by three points or more. The grades, every rating with its rater's note and these agreement figures are released with the paper.

The grade follows the PCR class (Table 2). Newcastle disease and coccidiosis droppings look abnormal, with mean grades 4.6 and 5.6, while salmonella droppings are close to healthy ones, 8.0 against 8.6, and nearly half of them look normal. The raters scored salmonella droppings as looser (mean form penalty 0.61 against 0.28) and with more undigested feed (0.44 against 0.09).

We expected the models to fail most on infected birds whose droppings look normal. They do not. Over the 899 PCR-positive photographs, the detector is right on 83% of the 190 that look normal and on 73% of the 709 that do not, and MobileNetV2 on 87% and 76%. The exception is

Table 2: Appearance grade and model accuracy on the lab-confirmed photographs. Normal looking means grade 9 or 10. Detector is our YOLO11n, MobileNetV2 is the seed 0 run. Accuracy is the share whose top class matches PCR.

PCR class	n	Mean grade	Normal looking	Acc. on normal looking	
				Detector	MobileNetV2
Healthy	347	8.6	62%	84%	66%
Salmonella	347	8.0	47%	93%	98%
Coccidiosis	366	5.6	7%	23%	23%
Newcastle	186	4.6	1%	1 photograph	

coccidiosis, where both models are right on only 6 of the 26 normal looking photographs. The high score on normal looking positives comes from salmonella, which both models call correctly 93 to 98% of the time even when the raters saw nothing abnormal. Part of that is a lean towards salmonella, since MobileNetV2 also calls 72 of the 215 normal looking healthy photographs salmonella (33%), and the detector 32 (15%). Telling a normal looking salmonella dropping from a healthy one by eye is hard, so cues other than the dropping's clinical signs, such as the surface, lighting or the camera used for one class, may be helping. These grades cannot show which.

4.6 Pen-scale detection

The close-up detector does not work at pen scale (Figure 7). On the held-out pen cameras it finds 29% of droppings, only 11% of its boxes are droppings, and it places 3.55 boxes on an average empty pen tile at the 0.25 threshold. After pen-scale fine-tuning, recall is 0.80

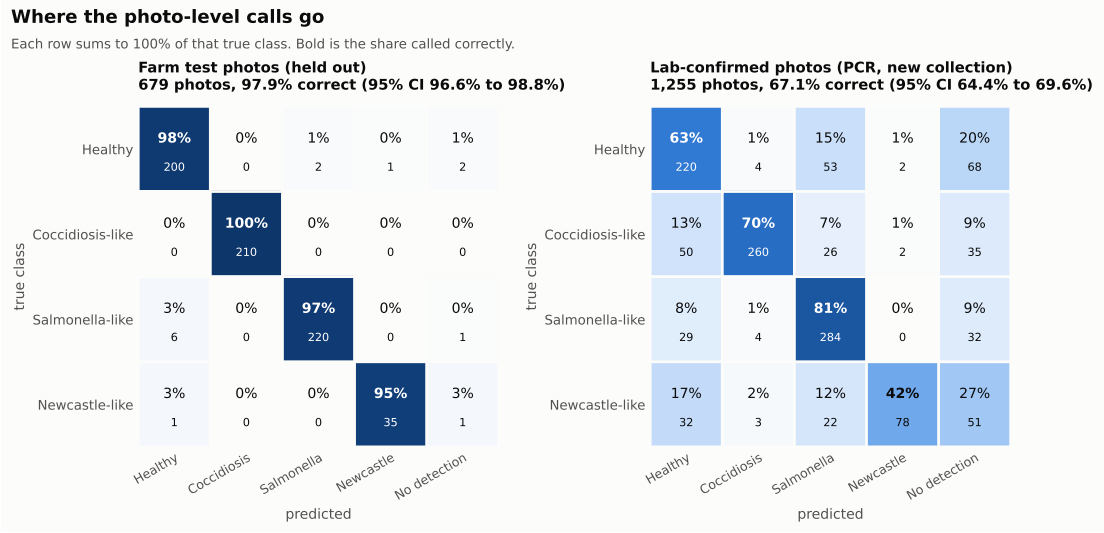


Figure 2: Detector calls per true class on the farm test (left) and the lab-confirmed photographs (right). Each row sums to 100%. Counts under each share.

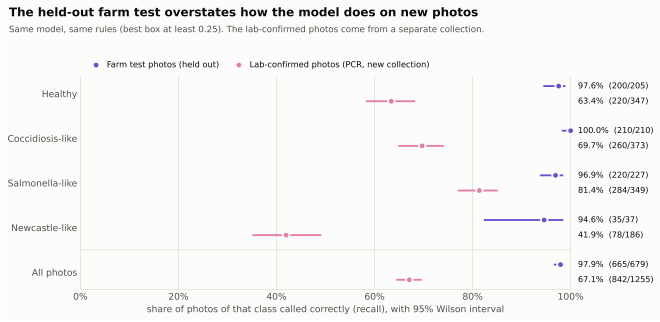


Figure 3: Per-class recall of the detector with 95% Wilson intervals, at the deployed 0.25 threshold.

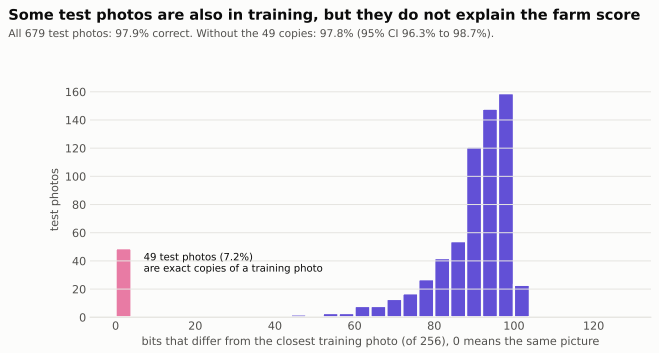


Figure 4: Hamming distance from each farm test photograph to its nearest training photograph (256-bit difference hash). The bar at zero holds the 49 exact copies.

and precision 0.93 on cameras the model never saw, and false boxes on empty tiles fall to 0.05 per tile. The close-up mAP50 changes from 0.873 to 0.853. Class accuracy at pen size remains weaker than detection (four-class mAP50 0.65). The pen model should therefore find a dropping, and a full-resolution patch or a close-up photograph should be judged by the close-up model.

Whole-photograph classifiers cannot be tested this way. They return one label for an image, and in a pen frame a dropping covers at most about 0.1% of the image. Locating the dropping is what makes the pen camera usable, and it is the main reason to use a detector in this setting rather than a better classifier.

4.7 The deployed pipeline

We also ran the pen model over a three-minute clip from the held-out PIO camera, into which ten test droppings were pasted at known times. Read at two frames a second with a short persistence rule, it flagged seven, five of them within two seconds of their appearance, and raised no false flags (Figure 8). This run was made on a laptop, and like the pen test it uses pasted droppings. On the Pi, the deployed pipeline proposes candidates with a still-spot finder and judges a full-resolution patch of each with the close-up detector. On the Raspberry Pi 5 the detector takes 168 ms at 320 pixels and 618 ms at 640 pixels with four threads, and the pen model 608 ms at 640 pixels. In the deployed pipeline the detector runs only on the candidate patches the spot finder flags, not on every frame. We did not time the baseline classifiers on the Pi. Native message passing between two sections takes 95 μs for a 57.6 KB frame and reply. When we killed any section with `SIGKILL`, the supervisor restarted it in 20 to 33 ms. Killing the spot finder cost the camera one frame and the other sections none.

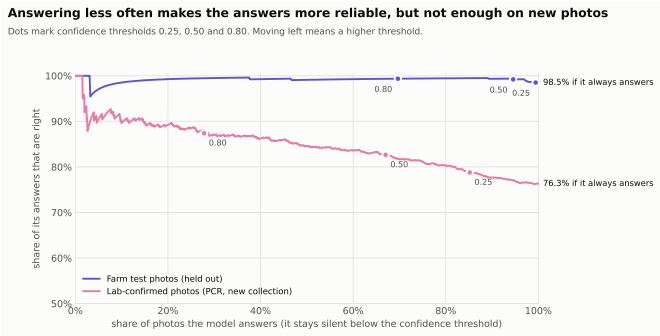


Figure 5: Accuracy of the detector's answers against the share of photographs it answers, as the confidence threshold rises. Dots mark the thresholds 0.25, 0.50 and 0.80.

5 Discussion

What the farm test measures. Every model we trained, including the two that earlier work recommended, reaches 98 to 99% on the random farm test and loses more than 20 points on photographs from a second collection of the same group. The original authors did not expect distribution shift to matter [25]. Our results show that it matters more than the choice of network. We cannot say from these data which difference between the collections causes the drop. The lab-confirmed photographs differ in label source, in when and how they were taken, and possibly in the farms they came from. Two observations narrow it down. First, exact duplicates are not the cause, because removing them leaves the farm score unchanged. Second, the errors are mostly missed detections and calls of healthy or salmonella-like, not swaps between the diseases. A change in how the droppings were photographed is consistent with this. Healthy droppings in the farm collection fill a median of 2.6% of the photograph, against 12 to 22% for the other classes, so photograph framing could act as a shortcut [13]. We have not tested that directly. Bodempudi et al. also found a drop across regions for a binary *Salmonella* task [6]. Together the two results suggest that published accuracies on this task should be read as within-collection accuracies.

Why use a detector. On photographs our detector is not more accurate than a whole-image classifier. Its value is that it says where the dropping is. That is what makes a fixed pen camera usable, because a dropping covers at most about 0.1% of a pen frame, and it is also what lets the explanation step measure the colour of the dropping itself rather than the litter around it. The pen-scale fine-tune shows that the detector can learn the size and setting of a pen camera from public video and pasted droppings alone, without a single labelled pen dropping. It lost 0.02 mAP50 on close-ups in the process.

Why a microkernel. A monitor in a barn runs unattended, and a crash in a camera driver or a model runtime should cost a few frames, not the device. On QNX each stage of our pipeline is an ordinary process, and the supervisor brought any killed stage back in 20 to 33 ms while the others kept running. The same structure is possible on Linux with process supervision, but on QNX the drivers, the file system and the network stack are also user processes that can fail and restart the same way [3]. We did not compare fault recovery against Linux on the same board, so we do not claim QNX recovers faster. Running the model on QNX also needed one fix in the platform's packages. The Python binding of ONNX Runtime in the QNX open-source repository does not load, because its build script treats QNX as Windows and looks for the wrong extension module. We worked around it with a 111-line C bridge to the runtime's C library and have prepared a patch for the package recipe.

Limitations. All photographs come from one region of Tanzania and one research group. The farm labels describe appearance and farm, and only the second collection was confirmed by PCR, so we treat the classes as screening categories, not diagnoses. The farm split is random by photograph because the release carries no farm or visit information. The pen test and the test clip use real droppings pasted onto real barn frames, so they measure whether a pen-sized dropping can be found on unseen litter, not field accuracy on naturally occurring droppings. We have no labelled droppings from a real pen camera. The detector and each baseline were trained once, with one seed and one set of hyperparameters that we did not tune, and EfficientNet-B0 for only 8 epochs. Run-to-run variation is therefore not measured, although it is small next to a drop of more than 20 points. Finally, the edge timings are for inference only and do not include image decoding, post-processing or power draw. The appearance grades come from three AI models following a rubric, not from veterinarians. Their agreement shows the grades are consistent, not that they are right, and three models from one vendor may share blind spots. The rubric is new and has not been validated against expert grading.

Future work. The most useful next step is labelled droppings from fixed pen cameras on more than one farm, split by farm. Short of that, the farm collection could be re-split with near-duplicate removal before splitting, and the size shortcut could be tested by cropping every class to the same framing. The box score needs calibration before it is shown to a farmer, and the 0.25 threshold should be chosen from a risk-coverage curve on validation data rather than fixed.

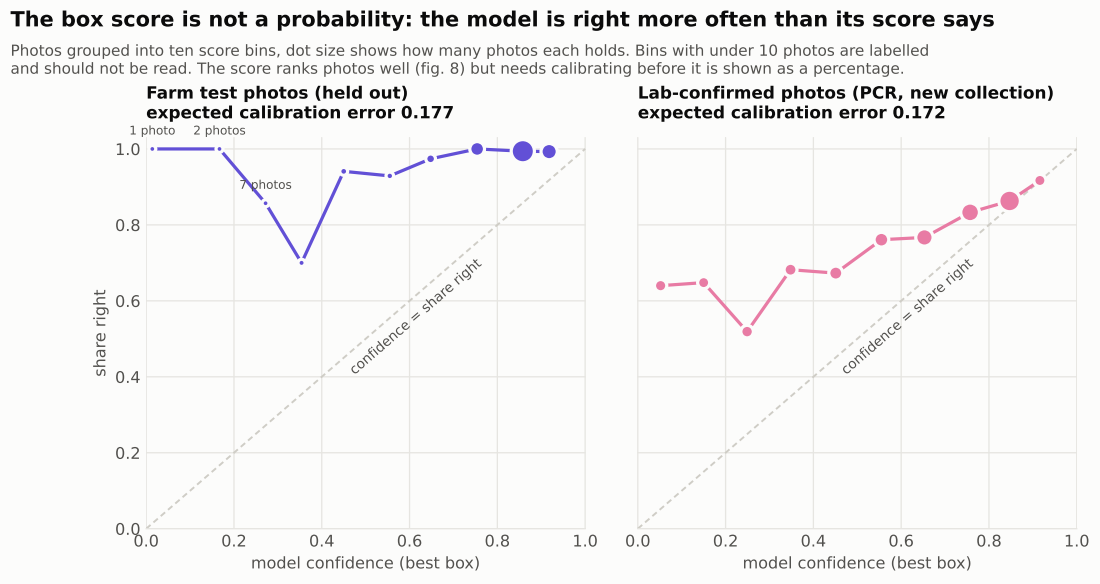


Figure 6: Reliability diagrams over ten score bins. Dot size shows the number of photographs in a bin. Bins with fewer than ten photographs are labelled and should not be read.

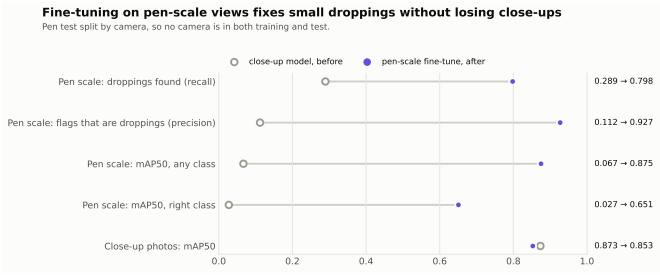


Figure 7: Pen-scale test on held-out cameras before (close-up model) and after fine-tuning. The pen test uses real droppings pasted onto real litter.

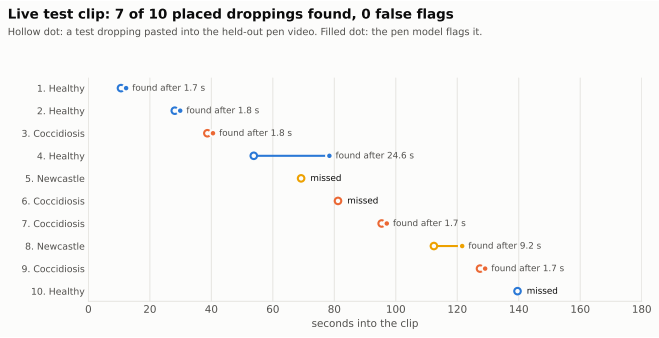


Figure 8: Test clip with an answer key. Hollow dots mark when each pasted dropping appears in the held-out video, filled dots when the pen model flagged it.

6 Conclusion

Classifiers of poultry droppings that score 98 to 99% on a random split of the Tanzanian farm photographs score 75 to 78% on lab-confirmed photographs from a second collection, and this holds for every architecture we tried, including our detector. We therefore recommend that future work on this task report a second collection alongside the farm test. A detector cannot fix this drop, but it is what makes a fixed pen camera usable. Fine-tuned on public barn video with pasted droppings, our YOLO11n detector finds 80% of pen-sized droppings on unseen cameras with 93% precision. The full pipeline runs on a Raspberry Pi 5 under QNX, where any failed stage restarts in tens of milliseconds while the rest keeps running.

Data and code availability

The farm and lab-confirmed photographs are Zenodo records 4628934 and 5801834, the PIO and NESTLER videos Zenodo records 16686320 and 15063494, all under CC BY 4.0. Our code, trained weights, per-photograph predictions, duplicate hashes, appearance grades with every rating, training logs and every figure with the table behind it are at [TODO: repository URL]. The figures are produced by `data-visualization/evaluation.py`, the baselines by `training/benchmark_classifiers.py`, and the image audit by `data-visualization/audit_images.py`.

Ethics

This study used only published photographs and videos. No animals were handled and no new animal data were collected.

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